

Towards knowledge-based networks

P Willis and I G Dufour

Intelligence in the telecommunications networks of the future will be formed from the interaction of intelligent terminals and core network intelligence. The intelligence will come from computing power which will at last become capable of exhibiting some elementary human characteristics. The network will possess knowledge of the user — notably location knowledge — and knowledge of the information available within it and may one day more properly be called a knowledge-based network.

1. Introduction

The convergence of computers and communications is producing many changes in society. The technology driving these changes will be a network of machines processing information. This network will appear to possess knowledge since it will be delivering the correct information to the correct people using both knowledge of the information and knowledge of the people and hence it might be described as a knowledge-based network.

The path to this future vision is influenced by three technological mega-trends summarised as bandwidth, intelligence and information, although, as will be shown, these terms need some qualification. To set them in context it is first necessary to review the pre-convergence history of telecommunications and computing technologies.

2. Pre-convergence history — telecommunications, LAN technologies and computing

The key trends in telecommunications developments over the last two or three decades have been the use of computers to control switches, the dominance of digital techniques, the availability of huge bandwidths via optical fibres in the fixed network and an increasing use of cellular radio networks which themselves are highly dependent on computer control. Asynchronous transfer mode (ATM) switches are starting to appear, bringing the traditionally speech-dominated telco world and the data-communications world together and providing an infrastructure for broadband multimedia services.

Key trends in local area network (LAN) technologies are increasing bandwidth, increasing efforts to interconnect LANs — more businesses are using enterprise networks which are formed from the interconnection of many LANs — and an increasing use of standards to allow for multivendor environments. The term ‘information network’ was often applied to these enterprise networks because information technology was applied to office automation, but the term is now taking on a very different meaning as will be discussed later.

In computing, the key trends have been ever-increasing speeds of operation, the development of large, reliable, centralised, mainframe computers, the development of the personal computer (PC) and UNIX-based workstations, and most recently the development of client/server architectures allowing them to work together. What has been seen is the emergence of an architecture where typically the data resides on a mainframe, processing is carried out on a server and user presentation is via a PC. Now further developments are being seen where processing power and data can be widely distributed — a concept that maps particularly well with distributed intelligence in networks [1]. The impact of microprocessors on customer premises equipment (CPE) is creating a range of intelligent appliances which can be connected to, and can co-operate with, the core telecommunications network, introducing the concept of co-operative intelligence.

3. Bandwidth, intelligence and information

Bandwidth, intelligence and information are three major areas influencing the development of knowledge-based networks and each is reviewed in the following sections.

3.1 *Bandwidth*

Bandwidth is an analogue term traditionally used to indicate the number of hertz required to carry a particular signal. It is ultimately an expression of information-handling capacity. That is why it is also now used as a synonym for data rates in digital systems. High bit rate systems are said to be of high bandwidth. Bandwidth is used here in its widest sense, as are its counterparts narrowband and broadband.

A key development affecting bandwidth trends has been the invention and relatively rapid deployment of optical fibres. The cost per bit transported has reduced significantly and yet currently only a fraction of the optical fibre's capacity is accessible [2]. Another key trend has been the ability to exploit copper wires for greater bandwidth [3]. At the same time compression technologies allow high bit rate source material to make efficient use of available bandwidth. Bit rates can also be traded with time in conjunction with digital storage capability. A film can be sent over a narrowband medium overnight, or over a broadband link in seconds — in each case the material is stored at the receiving end for watching as required.

3.2 *Intelligence*

The most conventional telecommunications view of an intelligent network (IN) as defined by ITU-T is where service control is separated from the network routing logic [4]. Another view is that it is simply about the convergence of telecommunications and computing. A further view of an intelligent network is of a humanised network embodying knowledge about the calling and the called parties and providing appropriate voice responses to voice inputs. The simplest way to imagine this is to think of the manual switchboard operator of a hundred years ago, i.e. the intelligence, knowledge and information held by the operator is replaced by data and logic embedded in computer programmes. These concepts can be extended to embrace non-speech and broadband networks. This view demonstrates that the use of the word intelligence in IN is something of a misnomer. Nevertheless, machine intelligence, as postulated by Alan Turing as long ago as 1950, is now becoming a probability rather than a distant possibility. It can be seen, therefore, that there is considerable ambiguity over what is meant by intelligence.

Today, the services that a so-called intelligent network enables are those such as number translation services (freefone, premium rate), virtual private networks (VPNs), cashless calling (chargecard) and cellular radio services. In the medium term these are being modified for a greater degree of tailoring to meet specific customer requirements. Examples of this are time-of-day routing variations (advanced number translation services) and call delivery determined by caller location (e.g. a national store chain directing calls to the branch nearest to the caller). In addition, capabilities such as number portability and personal numbering will appear, as will non-speech services and eventually broadband and bandwidth-on-demand services. At this stage the network is smart but it is unintelligent in human terms. It is a processing machine with flexible support systems and with a telecommunications interface to the customer. The computing capability is distributed and the network comprehends total mobility.

Beyond that, the telecommunications network may have gained enough intelligence through service control computing power to move on to the next stage. Here, core-network intelligence will work co-operatively with ever more intelligent terminal equipment, where both use machine intelligence to respond to the user as an individual comprehending context, emotion and other human characteristics. The network will thus be able to determine its actions in response to a wide range of inputs. It is at this point that the network can be better described as a knowledge-based network. Human to network communication for control purposes will almost certainly be dominated by speech even for controlling non-speech services. It will be in an environment where numerous service providers produce information or entertainment content ranging from digital libraries to video databases.

3.3 *Information*

For some time we have heard that we are now in the information age. This is usually taken to mean that we are passing from the industrial society to the post-industrial society. Put another way it means that service industries are becoming more important in some ways, not least in numbers of people employed, than manufacturing industries. In this context, people employed in 'service industries' certainly includes people working in hamburger shops, but it also means teachers, lawyers, management consultants, computer programmers and endless other occupations where people generate or use information for their work. Some, such as Peters [5], take the view that the 'service' content even in manufacturing industries is high. What matters is that a growing band of information intensive workers need to access and exchange information via telecommunications and the infrastructure to realise this will revolutionise telecommunications itself. The network to support this requirement is increasingly known as an

information network which embraces the traditional telecommunications network, the terminal appliances and invariably the information content as well.

A view of the impact of this revolution in telecommunications can be found in Naisbitt [6]. In this he defines telecommunications as encompassing telephones, televisions, computers and consumer electronics and regards it as the driving force that is simultaneously creating the huge global economy and making its parts smaller and more powerful. He notes that 'in the global economic network of the 21st century, information technology will drive change just as surely as manufacturing drove change in the industrial era.' This is taken a stage further by Negroponte [7] who distinguishes between an existing world order based on atoms (i.e. tangible products) and a new emerging one based on bits (i.e. information and entertainment in digital form).

4. Knowledge

The use of the word 'information' in the above section can frequently be substituted with the word 'knowledge'. While definitions of these terms are notoriously difficult, for convenience knowledge is assumed here to imply information with meaning. We are moving towards knowledge workers dominating a knowledge-based economy, both of which are highly dependent on telecommunications. The social and political implications of this movement were set out some years ago by Toffler [8]. His book includes a chapter on 'extra-intelligence' where 'we are looking towards networks.....(which).....act on the outside world, adding 'extra-intelligence' to the messages flowing through them' and a chapter on 'net power' that notes that 'in the super-symbolic economy, it is knowledge about knowledge that counts most.' The latter point has important implications for the design of networks if they are to support knowledge-based workers. Negroponte [7] takes this aspect substantially further forward and extends the principle to the general day-to-day requirements of individuals and the impact machine intelligence can have on them and their surroundings.

5. The birth of knowledge-based networks

The problem with identifying intelligence in an intelligent network is, as has been shown, due to the ambiguity of the term intelligence. It is more reasonable to identify where knowledge is located. A knowledge-based network is easier to understand in terms of what it does and what it offers the consumer. The consumer can buy intelligence in the form of processing power and software at lower cost than a telco can sell it in the form of network intelligence. However, the consumer may look more favourably on paying for knowledge obtained from a knowledge-based network because consumers understand that they cannot buy this knowledge cheaply. Today's intelligent networks have

limited knowledge, mainly knowledge of numbers, time of day and location, rather than being a knowledge-based machine using inferencing on the raw information.

5.1 The technology of knowledge-based networks

The convergence of telecommunications and computers to form knowledge-based networks is happening from several directions:

- computers are being used widely within telecommunications systems, e.g. modern telephone exchanges can be described as specialist computers,
- general-purpose computers are being used for additional control of telecommunications, e.g. intelligent networks [4],
- all communications can now be represented digitally (e.g. voice and video have been transported digitally for many years) — this has been the main driver in the multimedia revolution because all information and entertainment content can be represented as a stream of bits,
- ATM is seen as key to providing a single network infrastructure capable of supporting all forms of media,
- computer systems have been using data communications for decades (terminals connected to remote mainframes), but, recently, computers have had interfaces added to them to allow them to control non-data telephony calls directly [9] — from the computer's point of view the telephone has become a peripheral,
- ISDN eliminates the different interfaces to voice and data networks by providing a digital interface,
- the increasing size and interconnection of enterprise networks, and public data networks such as the Internet, means that the data network is increasingly beginning to act like telco networks, i.e. a computer user can communicate with any other computer, or computer user, in much the same way as a telephone user can contact any other telephone user.

Beyond this there is no consensus between the computer and communications industries over what form the network will take and it is most likely that it will be driven by the equipment that consumers buy. While a broadband infrastructure is required for many new services, the range of services possible over an essentially narrowband infrastructure is still large, partly because bit rate and time can be interchanged for entertainment services, and partly because a knowledge-based network will select and minimise essential information.

The future form of knowledge-based networks will also be influenced by non-technological factors such as:

- the development of standards for applications, peripherals, interfaces and design tools,
- the cost,
- the deployment of intelligent networks and services such as Internet by computer and communications companies,
- regulation.

5.2 *The 'location of knowledge' debate*

The computer, communications, information appliance, media and entertainment industries are currently competing to dominate the converging market-place which will, in practice, be dominated by those who have the most value to add. The most value will be added by those who own and control the knowledge, and the knowledge-based networks, in the converged market-place.

What is a network? The answer to this question will be influenced by whether you are a telecommunications engineer, a computer scientist, a psychologist or a politician. In general, a network is defined as a set of entities and the relationships between them, which have been formed for a co-operative purpose. The numerous views of a network are shown in Fig 1 which bears some resemblance to the ISO OSI 7-layer model. However, the intention of Fig 1 is to demonstrate the views people have of networks and not to present a view of their technical composition.

In the centre of Fig 1 is a circle representing the physical world. To the people in the physical world a network is made up of cables, fibres and telephone poles. Such people only think in terms of bandwidth — to them a national information infrastructure is copper or fibre in the ground with no consideration of the switching equipment and CPE required to use the bandwidth. This physical network has its own method of labelling, or addressing, the cables and fibres.

In the next layer are traditional telco people. These telco people take the cable infrastructure for granted and think that the network is formed by a collection of telephone exchanges (or switches). To the computer world the telco network provides end-to-end bit pipes. The telco network uses telephone numbers to address end connections.

At the network layer, or the Internet layer, reside routers and CPE. These have unique network addresses. At this layer the network is formed by a collection of subnets, routers and hosts (end-systems). The network is a logical

abstraction of the physical and telco networks (or bit-pipes). Network layer people think in terms of offering logical connectivity between machines. Traditional IN functionality could be considered to reside at the network layer as it deals with the routing of telephone connections.

At the transport layer are services provided by operating systems, accessed via the APIs (application programming interfaces) offered by the operating systems. This is a software world used by application writers. Variants of APIs exist, some of which are over 20 years old and programmers tend to use their favourite, a popular one being 'sockets'. To programmers the network looks like various transport services between machines, literally a socket offering a transport service, e.g. connectionless data, to other machines. Programmers need no knowledge of the details of the Internet, the telco network or the physical network.

At the applications layer sit the applications that people use. The applications are written by programmers who use the services of the transport layer. Applications present a wide range of interfaces to users, ranging from applications where the user has to be network-address aware or telephone-number aware to applications where no knowledge of a target connection is required. In the Internet this application layer is often called Cyber-space. Most people have a limited knowledge of the area of Cyber-space they use and usually no knowledge of the lower network layers is required. The preferred addressing mechanisms in Cyber-space are addresses with which people can associate, for example names of people.

The ultimate network is that formed by the people that are using the information network. The term Cyber-nation has been used by people in the Internet to identify the people that use it. The Cyber-nation is essentially a nation without geography. When politicians discuss the information superhighway or the global information infrastructure they are invariably concerned about the impact of the information network on society (and their own future). Globally available bits threaten the existence of the nation state but no one yet knows what might replace it although it is already postulated that the number of 'countries' may increase [6].

6. **The era of knowledge-based networks**

As has been outlined above the future will increasingly be dominated by intelligent telecommunications networks and intelligent terminals. These will jointly sustain a wide range of applications involving many information and entertainment types provided by independent service providers. The future should thus bring about

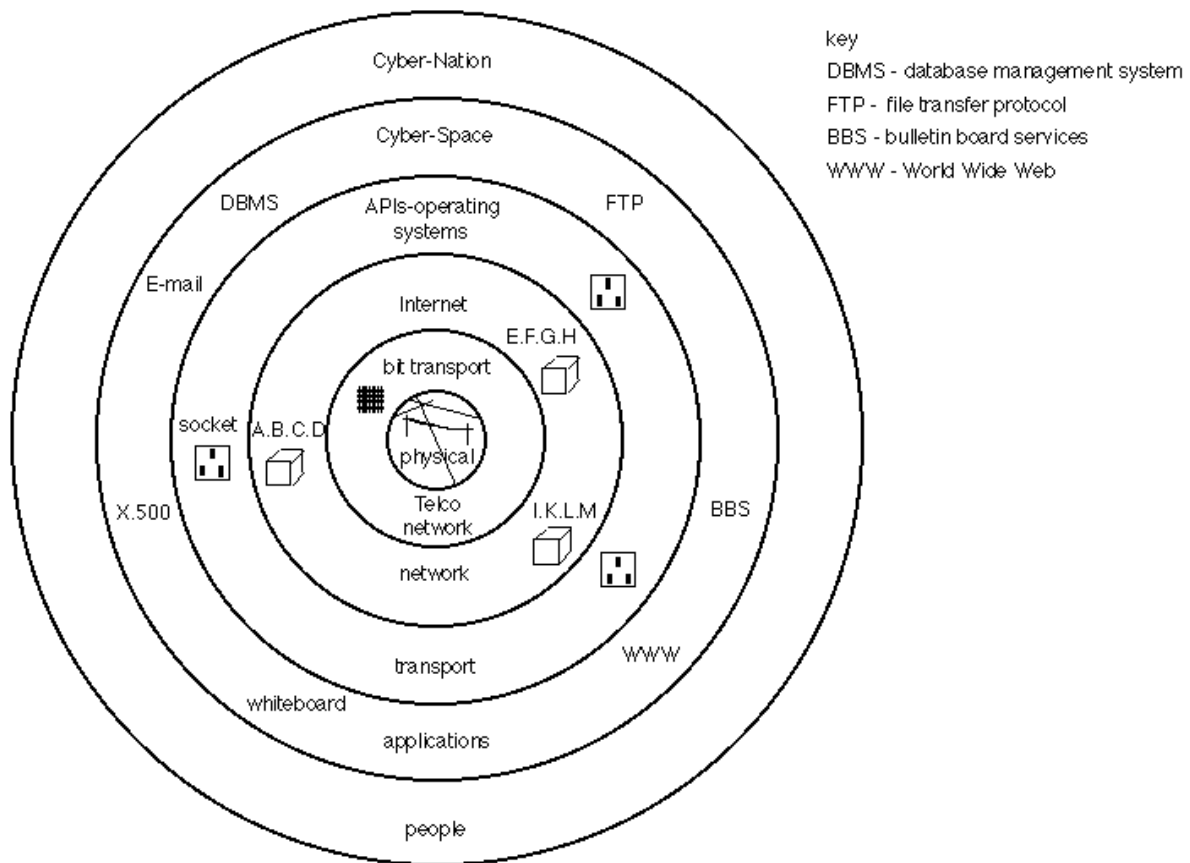


Fig 1 What is a network?

increased competition in the areas of information technology and communications. There will be several communications networks using slightly different network technologies forming knowledge-based networks just as roads, rail and air provide a variety of transport systems. There will also be a range of intelligent terminals, such as smart TVs, multimedia phones, multimedia PCs. These intelligent terminals will have more technological similarities than dissimilarities. They may be specialised boxes, not general-purpose machines like the computers of today, because they will offer ease-of-use for very particular information retrieval and processing functions. Every home will have several and they may communicate with each other as well as with core networks — they will be information appliances.

Information appliances have been with us for some time. The pocket calculator is an example of an information appliance. It does just one job but it does it very well and it is easy to use. If any domestic appliance is considered, a cooker, a television, a kettle, they all perform a limited function very well. Appliances are easy to install and use. Plug them into the required services, e.g. gas or electricity, and turn on. General-purpose computers as used for most of today's information processing are not appliances — a general-purpose computer is not a simple-to-use device that

performs one task well. In the future computers will still be used but they will be highly specialised, performing a limited range of tasks very well. The information appliance will be a plug-in and turn-on box that will process information in a certain well-defined manner. For example, an information appliance might be a knowledge-retrieval system, it might be the size of a button and use only an audio interface, and could connect into global knowledge-based networks via wireless links.

The suppliers of multimedia services information will become utilities. The information appliances will plug into utilities like today's domestic appliances plug into the gas, communications or electricity utilities. We will depend on information utilities just as we depend on electricity utilities today. The loss of an information utility in the future will have as grave an impact as the loss of an electricity utility today.

The form of communications networks is harder to predict. However, imagine a world with three major types of providers of communications networks:

- interactive cable TV networks,
- multimedia telephony networks,
- packet data networks.

These networks will interconnect for a range of reasons, including customers seeing added value in being able to obtain information from suppliers residing on other networks. Also, businesses will want to communicate with everyone regardless of their network connectivity. There should be no discrimination between consumers and providers — anyone will be able to be a seller of information as well as a consumer, although in practice information providers may have different bandwidth and service requirements from consumers. Interactive cable TV, the multimedia telephone and the multimedia workstation are all versions of very similar information appliances. The main difference in these information appliances is that they are marketed differently. Some information appliances may be capable of obtaining their communications services from different types of communications networks. An information network is not complete without the appliances, or the people that are providing and consuming the information. The various communications networks may interconnect via some backbone networks and these may even be global backbone networks. Some networks will have peer arrangements with other networks and set up their own interconnection. Mobile information appliances will be common. People will use portable PCs, organisers and personal intelligent communicators [10] which will connect into the global network using wireless techniques.

Information consumers and providers with their information appliances supported by the information utilities will form the tele-information market-place. Locating where intelligence and knowledge is in this complex web is not straightforward. Although all the information appliances have similar processing power they are different, for example:

- the multimedia workstation connected to the Internet has considerable intelligence whilst the network has small amounts of intelligence (a little knowledge exists in the Internet to provide routeing and naming information),
- the multimedia telephone may be un-intelligent with most of the intelligence provided by the knowledge-based network.
- interactive cable-TV set-top boxes may be somewhere between the intelligence of the workstation and the telephone, with content and knowledge in the service provider's domain.

A knowledge-based network has the possibility of bringing the consumer closer to the knowledge in a more efficient manner than in the Internet. One of several problems with the Internet is finding where the knowledge is located whereas this is not a problem with a knowledge-based network.

7. Conclusions

The essence of convergence is digital technology — telecommunications, datacommunications, computing, media content and applications are all digital. Communications networks are becoming closely tied to computing capability forming a distributed computing network. The network operators of the future will therefore be operating a huge distributed computing network. This network will process information by applying intelligence to it in the manner understood by machine intelligence rather than that currently understood by IN. The network will appear to its customers to have knowledge, it will appear to provide and operate on knowledge both of the people and of the information, and it will therefore be a knowledge-based network. The knowledge-based network will come to fruition in an era when the economic, social and political structures will be dominated by knowledge and information within a digital environment and in this future converged world there will be many players with constantly changing boundaries between them.

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Peter Willis joined BT Laboratories in 1988 and has worked mainly in datacommunications including experience in SMDS, ISDN, X.25, TCP/IP and OSI network management. He started work in the area of network intelligence in April 1993 where he brought a different view-point, that of datacommunications rather than the more usual background of voice communications. He recently developed and helped provide BT's public Internet access service and in November 1994 he moved to managed network services to manage the service.



Ian Dufour started his career with BT as an apprentice, and his early experiences covered analogue and digital transmission systems and computerised planning support systems. He introduced single-mode optical cables and systems in junction and local networks and also implemented blue-book standard ISDN in the BT network. He moved to BT Laboratories in 1991, first setting up a network integration facility and then establishing a team working on network intelligences. He is a European Engineer, a Chartered Engineer and a Fellow of the IEE.